Predicting Term-Deposit Subscription From Bank Telemarketing Data

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Overview

This report describes a model that predicts, before a marketing call is made, whether a client of a Portuguese bank will subscribe to a term deposit. The problem type is supervised learning – binary classification, with the target *y* (yes or no). The data are the Bank Marketing dataset from the UCI Machine Learning Repository (https://archive.ics.uci.edu/dataset/222/bank+marketing), and the main model is a class-weighted random forest that is compared with a majority-class dummy and a logistic regression baseline. On held-out clients, the forest ranks likely subscribers about 3.9 times better than a random call list (PR-AUC 0.454 against a baseline of 0.117).

Dataset Description

The data record 45,211 clients of a Portuguese bank who were contacted by phone in direct-marketing campaigns between May 2008 and November 2010 (Moro et al., 2014). The file bank-full.csv has 17 columns: 16 input features and the target *y*, which records whether the client subscribed to a term deposit. The features describe the client (age, job, marital status, education, credit default, average yearly balance, housing loan and personal loan), the last contact of the current campaign (contact type, day, month, call duration and number of contacts) and the previous campaign (days since the last contact, number of earlier contacts and the earlier outcome). Only 5,289 clients (11.7%) subscribed, so the classes are strongly imbalanced. The file has no missing cells and no duplicate rows, but the placeholder “unknown” fills 81.8% of the previous-outcome values and 28.8% of the contact-type values. The dataset is public under a CC BY 4.0 license.

Modeling Approach

The call-duration column was removed before modeling. Moro et al. (2014) note that call duration strongly affects the outcome but is only known after the call has ended, so it must be excluded from a realistic predictive model. A binary flag for previously contacted clients was added, because the days-since-last-contact feature uses −1 as a code for "never contacted" (81.7% of clients) rather than a real number of days. The data were then split into a stratified training set (36,168 clients) and test set (9,043 clients), and both sets keep the 11.7% subscription rate. All preprocessing (imputation, one-hot encoding of the 9 categorical features and standardization of the 7 numeric features, giving 51 model inputs) was placed inside a scikit-learn pipeline. The pipeline learns every statistic from training data only, including inside each cross-validation fold, which is the recommended defense against data leakage (scikit-learn developers, 2026). The "unknown" labels were kept as their own category, because a missing record of a previous outcome is informative in itself.

A random forest was chosen as the main model because the inputs mix numeric and categorical features, and the useful signals, such as call timing and the outcome of earlier campaigns, may interact in non-linear ways. Single decision trees handle such inputs with little preparation but overfit and change a great deal with small changes in the data. A random forest reduces this variance by averaging many trees, each grown on a bootstrap sample with a random subset of features considered at each split, at the cost of a slight increase in bias (scikit-learn developers, 2026). Both the forest and the logistic regression weighted the classes in inverse proportion to their frequency, so the rare subscribers were not ignored. The forest's depth and minimum leaf size were tuned with a 4-setting grid search over 5-fold stratified cross-validation (300 trees). The best setting used unlimited depth and at least 5 clients per leaf. The pipeline

structure and the permutation-importance procedure were adapted from examples in the scikit-learn user guide.

Accuracy is not a useful headline metric here, because always answering “no” is already 88.3% accurate. The evaluation therefore centers on precision (the share of flagged clients who subscribe), recall (the share of subscribers who are flagged), their harmonic mean, F1, and the area under the precision–recall curve (PR-AUC, computed as average precision). Saito and Rehmsmeier (2015) show that when positives are rare, precision–recall analysis describes a classifier's practical performance more faithfully than the ROC curve, whose false-positive rate is diluted by the large number of negatives. ROC-AUC is still reported for comparison. PR-AUC was also the score optimized during tuning, because the business task is to rank clients so that the likeliest subscribers are called first. The decision threshold (0.49) was chosen to maximize F1 on out-of-fold training predictions, so the test set played no part in it.

The random forest makes no assumption of linear relationships, normal distributions or equal feature scales, and tree-based models do not need normalized inputs (scikit-learn developers, 2026). Logistic regression, by contrast, models the log-odds of subscribing as a linear combination of the inputs, which limits how well it captures interactions. Both models assume that the clients they will score in the future come from the same population as the training clients. Mehrabi et al. (2021) describe two ways this assumption can fail: representation bias, when the sample does not reflect the target population, and temporal bias, when populations and behaviors change over time.

Results

Table 1 compares the models on the 9,043 held-out test clients. The dummy has the highest accuracy (0.883) but finds no subscribers at all. The random forest beats logistic

regression on both ranking metrics (PR-AUC 0.454 vs. 0.409; ROC-AUC 0.804 vs. 0.772) and on F1 (0.485 vs. 0.374). Logistic regression flags more clients, which gives it higher recall (0.624) but much lower precision (0.267). The forest's test PR-AUC is close to its cross-validated value (0.441 ± 0.017), so the estimate is stable. Moving the threshold from 0.50 to 0.49 changed little: recall rose by 0.019 and precision fell by 0.008.

Table 1

Test-Set Performance of the Baseline and Main Models

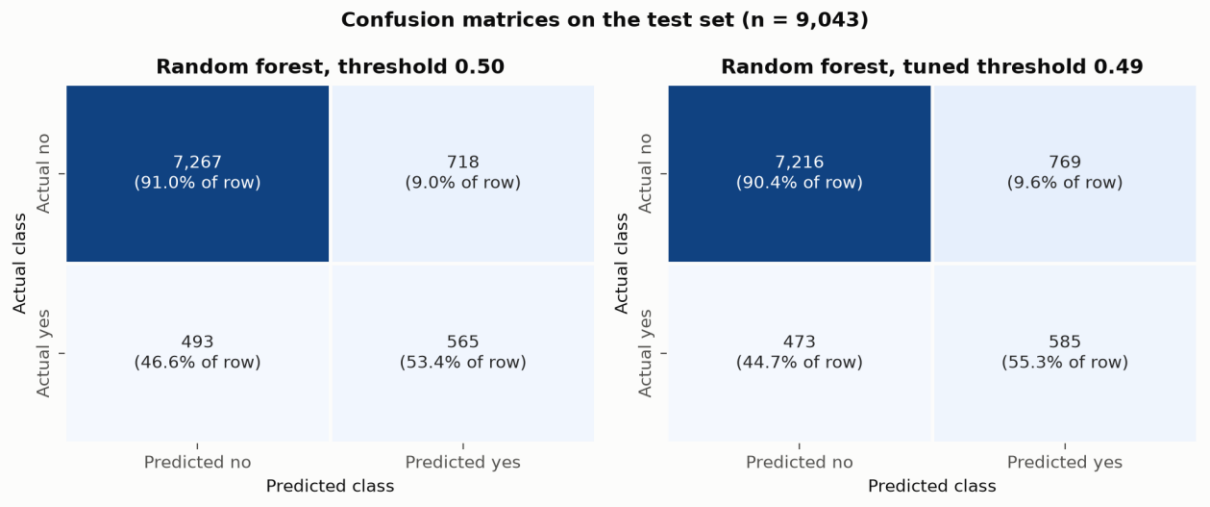
Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Dummy (majority class)	0.883	0.000	0.000	0.000	0.500	0.117
Logistic regression	0.755	0.267	0.624	0.374	0.772	0.409
Random forest, t = 0.50	0.866	0.440	0.534	0.483	0.804	0.454
Random forest, t = 0.49	0.863	0.432	0.553	0.485	0.804	0.454

Note. n = 9,043 test clients, of whom 11.7% subscribed. t = decision threshold; the tuned threshold was chosen on out-of-fold training predictions. PR-AUC = average precision; a random ranking scores the subscription rate (0.117).

Figure 1 shows that at the tuned threshold the forest flags 1,354 of the 9,043 test clients (15.0%), and 585 of them subscribe. It misses 473 of the 1,058 subscribers (44.7%) and wrongly flags 769 of the 7,985 non-subscribers (9.6%).

Figure 1

Confusion Matrices of the Random Forest on the Test Set

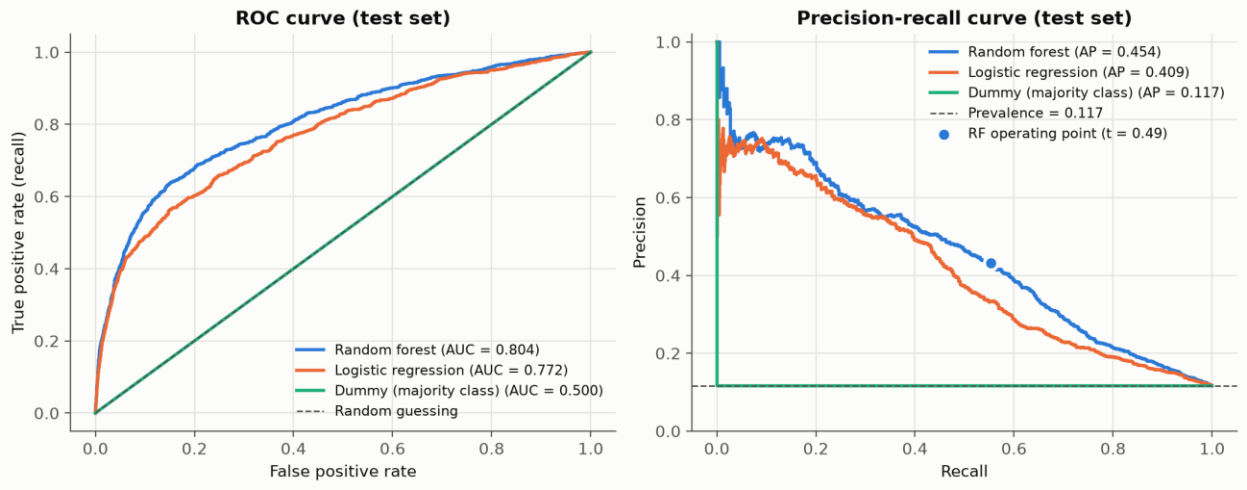


Note. Left: default threshold of 0.50. Right: tuned threshold. Percentages are shares of each actual class (row).

Figure 2 shows that the forest's ROC curve lies above that of logistic regression almost everywhere. The precision–recall panel shows the cost of finding more subscribers: precision falls steadily as recall rises, so no threshold gives high precision and high recall together. At the chosen operating point, a recall of 0.553 comes with a precision of 0.432, which means most flagged clients still do not subscribe even though the ROC-AUC of 0.804 looks strong.

Figure 2

ROC and Precision–Recall Curves of the Three Models on the Test Set

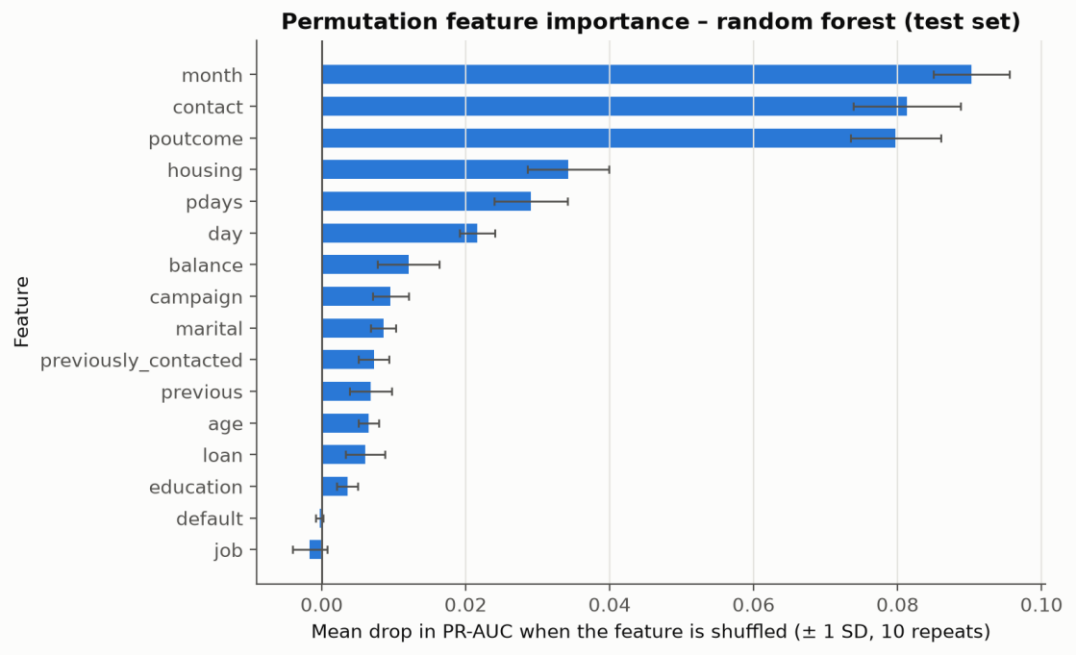


Note. The dashed lines show random guessing (ROC) and the subscription rate (precision–recall). The dot marks the random forest's tuned operating point.

Figure 3 shows that the forest relies mostly on the month of contact (mean drop in PR-AUC 0.090 when shuffled), the contact type (0.081) and the outcome of the previous campaign (0.080). Client characteristics such as age (0.007), education (0.004) and job (−0.002) add almost nothing. The model has therefore learned mainly when and how the bank called and how the client responded before, rather than who the client is.

Figure 3

Permutation Feature Importance of the Random Forest on the Test Set



Note. Importance is the mean drop in test-set PR-AUC when one original column is randomly shuffled, over 10 repeats; error bars show ±1 standard deviation.

Interpretation for a Non-Technical Audience

Imagine the bank has a list of 100 clients it could call. Without the model, about 12 of them would say yes. If the bank calls only the clients the model picks out, about 43 out of every 100 calls end with a new deposit, roughly 3.7 times as many as calling at random. The model is far from perfect: it still overlooks about 45 out of every 100 clients who would have said yes, and more than half of the clients it picks out (about 57 in 100) still say no.

The model is therefore best used to decide whom to call first, not to decide that some clients should never be called. Its most useful clues are the time of year of the call, whether the client was reached on a mobile phone or a landline (or the bank did not record which), and whether the client said yes to an earlier campaign. Personal details such as age or job matter

much less. One caution matters for planning: the model learned from calls made between 2008 and 2010, and when it was tested only on the most recent of those calls, it was clearly less reliable. It should be rechecked on new calls before the bank depends on it.

Limitations and Potential Bias

The most important limitation is that the headline results come from a random split of data that are ordered in time. When the same tuned model was trained on the earliest 80% of rows and tested on the latest 20%, the subscription rate rose from 6.7% to 31.6% between the two periods, and ROC-AUC fell from 0.804 to 0.682. PR-AUC on the later period (0.470) looks similar, but it is only 1.5 times its new baseline of 0.316, compared with 3.9 times under the random split. This matches what Mehrabi et al. (2021) call temporal bias, and it means Table 1 overstates how well the model would forecast a future campaign. The heavy reliance on month and day of contact in Figure 3 makes the model especially sensitive to changes in the campaign calendar.

A second limitation is variance. The forest scores a PR-AUC of 0.718 on its own training data against 0.441 in cross-validation, so its deep trees partly memorize the training clients. The most regularized setting in the grid narrowed this gap (training PR-AUC 0.497) but also lowered cross-validated PR-AUC to 0.431, so some extra variance was accepted in exchange for better ranking. The metrics themselves also have limits. The baseline of a precision–recall curve equals the share of positives (Saito & Rehmsmeier, 2015), so PR-AUC values cannot be compared directly across periods or datasets with different subscription rates. This is why the temporal check above is read against its own baseline.

A specific ethical concern is how the model treats clients of different ages. Table 2 shows that precision is similar across age bands (0.40 to 0.46), but recall is not: subscribers aged 60 or

over are flagged 92.9% of the time, compared with 46.1% for ages 30 to 44 and 47.3% for ages 45 to 59. The model flags 70.9% of all clients aged 60 or over, against 11.4% of those aged 30 to 44. Under the equal-opportunity criterion, which asks that truly positive people be selected at the same rate in every group (Mehrabi et al., 2021), the model disadvantages middle-aged subscribers. In practice, it would also direct a large share of sales calls toward older and retired clients, which raises the question of whether that group would face more pressure to buy financial products than others. The 60-and-over band contains only 337 test clients, so its estimates are less certain.

Table 2

Random Forest Test-Set Performance by Age Band at the Tuned Threshold

Age band	n	Subscription rate	Share flagged	Precision	Recall
18–29	1,003	17.8%	26.9%	0.459	0.693
30–44	4,823	10.0%	11.4%	0.405	0.461
45–59	2,880	9.8%	10.2%	0.456	0.473
60+	337	33.2%	70.9%	0.435	0.929

Note. Threshold t = 0.49. Subscription rate is the actual share of subscribers in each band.

One step was taken and one is proposed to reduce this risk. The subgroup audit in Table 2 was run before any use of the model, so the gap is documented rather than hidden. The proposed step is to make the audit a release gate: at every retraining on recent data, recall and the share flagged are recomputed for each age band, and flagged clients aged 60 or over receive a human suitability check and a cap on how often they can be called.

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